

Vektoranalysis-Kompetenz

Es seien $\mathbf{a}, \mathbf{r}, \mathbf{x} \in \mathbb{R}^3$, $r := \|\mathbf{r}\|$, $\mathbf{r} = (x, y, z)^T$ und $\nabla_r = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)^T$.

Hierbei sind $\mathbf{a}$ und $\mathbf{x}$ konstante Vektoren.

Man berechne:

$$\begin{aligned} & \vec{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad r = \|\vec{r}\| = \sqrt{x^2 + y^2 + z^2} \\ 1. \quad \nabla_r r &= 1. \quad \nabla_r r = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot \sqrt{x^2 + y^2 + z^2} = \begin{pmatrix} \frac{1}{2} \cdot \frac{1}{\sqrt{x^2 + y^2 + z^2}} \cdot 2x \\ \frac{1}{2} \cdot \frac{1}{\sqrt{x^2 + y^2 + z^2}} \cdot 2y \\ \frac{1}{2} \cdot \frac{1}{\sqrt{x^2 + y^2 + z^2}} \cdot 2z \end{pmatrix} = \frac{1}{r} \cdot \vec{r} = \underline{\underline{\vec{e}_r}} \\ 2. \quad \nabla_r \cdot \vec{r} &= 2. \quad \nabla_r \cdot \vec{r} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 1 + 1 + 1 = \underline{\underline{3}} \\ 3. \quad \nabla_r (\mathbf{a}^T \mathbf{r}) &= 3. \quad \nabla_r \cdot (\vec{a} \cdot \vec{r}) = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot \left[\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} \right] = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot (a_1 x + a_2 y + a_3 z) = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \underline{\underline{\vec{a}}} \\ 4. \quad \nabla_r \frac{1}{r} &= 4. \quad \nabla_r \cdot \frac{1}{r} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot \frac{1}{\sqrt{x^2 + y^2 + z^2}} = \begin{pmatrix} -\frac{1}{2} \cdot \frac{1}{r^3} \cdot 2x \\ -\frac{1}{2} \cdot \frac{1}{r^3} \cdot 2y \\ -\frac{1}{2} \cdot \frac{1}{r^3} \cdot 2z \end{pmatrix} = -\frac{1}{r^3} \cdot \vec{r} = -\frac{1}{r^2} \cdot \frac{1}{r} \cdot \vec{r} = -\frac{1}{r^2} \cdot \underline{\underline{\vec{e}_r}} \\ 5. \quad \nabla_r r^{-3} &= 5. \quad \nabla_r \cdot r^{-3} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot (x^2 + y^2 + z^2)^{-\frac{3}{2}} = \begin{pmatrix} -\frac{3}{2} \cdot \frac{1}{r^{\frac{5}{2}}} \cdot 2x \\ -\frac{3}{2} \cdot \frac{1}{r^{\frac{5}{2}}} \cdot 2y \\ -\frac{3}{2} \cdot \frac{1}{r^{\frac{5}{2}}} \cdot 2z \end{pmatrix} = -\frac{3}{r^{\frac{5}{2}}} \cdot \vec{r} = -\frac{3}{r^{\frac{5}{2}}} \cdot \frac{1}{r} \cdot \vec{r} = -\frac{3}{r^{\frac{7}{2}}} \cdot \underline{\underline{\vec{e}_r}} \\ 6. \quad \nabla_r \frac{\vec{a} \cdot \vec{r}}{r^3} &= 6. \quad \nabla_r \frac{\vec{a} \cdot \vec{r}}{r^3} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot \frac{1}{\sqrt{x^2 + y^2 + z^2}^3} \cdot \left[\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} \right] = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot \left[(a_1 x + a_2 y + a_3 z) \cdot (x^2 + y^2 + z^2)^{-\frac{3}{2}} \right] = \begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix} = \underline{\underline{\vec{a}}} \\ \Rightarrow A_1 &= a_1 \cdot (x^2 + y^2 + z^2)^{-\frac{3}{2}} + (a_1 x + a_2 y + a_3 z) \cdot -\frac{3}{2} \cdot (x^2 + y^2 + z^2)^{-\frac{5}{2}} \cdot 2x = \frac{a_1}{r^3} - \frac{3x(\vec{a} \cdot \vec{r})}{r^5} \\ \Rightarrow A_2 &= a_2 \cdot (x^2 + y^2 + z^2)^{-\frac{3}{2}} + (a_1 x + a_2 y + a_3 z) \cdot -\frac{3}{2} \cdot (x^2 + y^2 + z^2)^{-\frac{5}{2}} \cdot 2y = \frac{a_2}{r^3} - \frac{3y(\vec{a} \cdot \vec{r})}{r^5} \\ \Rightarrow A_3 &= a_3 \cdot (x^2 + y^2 + z^2)^{-\frac{3}{2}} + (a_1 x + a_2 y + a_3 z) \cdot -\frac{3}{2} \cdot (x^2 + y^2 + z^2)^{-\frac{5}{2}} \cdot 2z = \frac{a_3}{r^3} - \frac{3z(\vec{a} \cdot \vec{r})}{r^5} \\ \Rightarrow \underline{\underline{\vec{a}}} &= \frac{1}{r^3} \cdot \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} - \frac{3(\vec{a} \cdot \vec{r})}{r^5} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \underline{\underline{\vec{a}}} \\ 7. \quad \nabla_r (\vec{r} - \vec{x}) &= 7. \quad \nabla_r (\vec{r} - \vec{x}) = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot \begin{pmatrix} x - x_1 \\ y - x_2 \\ z - x_3 \end{pmatrix} = 1 + 1 + 1 = \underline{\underline{3}} \\ 8. \quad \nabla_r \cdot \|\vec{r} - \vec{x}\| &= 8. \quad \nabla_r \cdot \|\vec{r} - \vec{x}\| = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot \left\| \begin{pmatrix} x - x_1 \\ y - x_2 \\ z - x_3 \end{pmatrix} \right\| = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot \sqrt{(x - x_1)^2 + (y - x_2)^2 + (z - x_3)^2} = \\ &= \frac{1}{2} \cdot \frac{1}{\|\vec{r} - \vec{x}\|} \cdot \begin{pmatrix} 2x - 2x_1 \\ 2y - 2x_2 \\ 2z - 2x_3 \end{pmatrix} = \frac{\vec{r} - \vec{x}}{\|\vec{r} - \vec{x}\|^2} \\ 9. \quad \nabla_r \cdot \frac{1}{\|\vec{r} - \vec{x}\|^3} &= 9. \quad \nabla_r \cdot \frac{1}{\|\vec{r} - \vec{x}\|^3} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot \frac{1}{((x - x_1)^2 + (y - x_2)^2 + (z - x_3)^2)^{\frac{3}{2}}} = -\frac{1}{2} \cdot \frac{1}{((x - x_1)^2 + (y - x_2)^2 + (z - x_3)^2)^{\frac{3}{2}}} \cdot \begin{pmatrix} 2x - 2x_1 \\ 2y - 2x_2 \\ 2z - 2x_3 \end{pmatrix} = \\ &= -\frac{1}{\sqrt{(x - x_1)^2 + (y - x_2)^2 + (z - x_3)^2}^3} \cdot \begin{pmatrix} x - x_1 \\ y - x_2 \\ z - x_3 \end{pmatrix} = -\frac{\vec{r} - \vec{x}}{\|\vec{r} - \vec{x}\|^3} \end{aligned}$$

$$10.) \nabla_r \cdot \frac{1}{\|\vec{r}-\vec{x}\|^3} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot \frac{1}{((x-x_1)^2 + (y-x_2)^2 + (z-x_3)^2)^{\frac{3}{2}}} = -\frac{3}{2} \cdot \frac{1}{((x-x_1)^2 + (y-x_2)^2 + (z-x_3)^2)^{\frac{3}{2}}} \begin{pmatrix} 2x-2x_1 \\ 2y-2x_2 \\ 2z-2x_3 \end{pmatrix} =$$

$$= -\frac{3}{\sqrt{(x-x_1)^2 + (y-x_2)^2 + (z-x_3)^2}^{\frac{5}{2}}} \cdot \begin{pmatrix} x-x_1 \\ y-x_2 \\ z-x_3 \end{pmatrix} = -\frac{3 \cdot (\vec{r}-\vec{x})}{\|\vec{r}-\vec{x}\|^5}$$

$$11.) \nabla_r \times (\vec{r}-\vec{x}) = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \times \begin{pmatrix} x-x_1 \\ y-x_2 \\ z-x_3 \end{pmatrix} = \begin{pmatrix} 0-0 \\ 0-0 \\ 0-0 \end{pmatrix} = \underline{\underline{\vec{0}}}$$

$$12.) \nabla_r \times \frac{(\vec{r}-\vec{x})}{\|\vec{r}-\vec{x}\|^3} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \times \begin{pmatrix} \frac{x-x_1}{\|\vec{r}-\vec{x}\|^3} \\ \frac{y-x_2}{\|\vec{r}-\vec{x}\|^3} \\ \frac{z-x_3}{\|\vec{r}-\vec{x}\|^3} \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial y} \frac{z-x_3}{\|\vec{r}-\vec{x}\|^3} - \frac{\partial}{\partial z} \frac{y-x_2}{\|\vec{r}-\vec{x}\|^3} \\ \frac{\partial}{\partial z} \frac{x-x_1}{\|\vec{r}-\vec{x}\|^3} - \frac{\partial}{\partial x} \frac{z-x_3}{\|\vec{r}-\vec{x}\|^3} \\ \frac{\partial}{\partial x} \frac{y-x_2}{\|\vec{r}-\vec{x}\|^3} - \frac{\partial}{\partial y} \frac{x-x_1}{\|\vec{r}-\vec{x}\|^3} \end{pmatrix} = \begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix} = \underline{\underline{\vec{0}}}$$

$$A_1 = \frac{\partial}{\partial y} \left[(z-x_3) \cdot ((x-x_1)^2 + (y-x_2)^2 + (z-x_3)^2)^{-\frac{3}{2}} \right] - \frac{\partial}{\partial z} \left[(y-x_2) \cdot ((x-x_1)^2 + (y-x_2)^2 + (z-x_3)^2)^{-\frac{3}{2}} \right] =$$

$$= (z-x_3) \cdot \left(-\frac{3}{2}\right) \cdot \|\vec{x}-\vec{r}\|^{-5} \cdot (2y-2x_2) - (y-x_2) \cdot \left(-\frac{3}{2}\right) \cdot \|\vec{x}-\vec{r}\|^{-5} \cdot (2z-2x_3) =$$

$$= -3 \cdot \|\vec{x}-\vec{r}\|^{-5} \cdot \underbrace{[(z-x_3) \cdot (y-x_2) - (y-x_2) \cdot (z-x_3)]}_{\vec{0}} = \underline{\underline{0}}$$

$$A_2 = \frac{\partial}{\partial z} \left[(x-x_1) \cdot ((x-x_1)^2 + (y-x_2)^2 + (z-x_3)^2)^{-\frac{3}{2}} \right] - \frac{\partial}{\partial x} \left[(z-x_3) \cdot ((x-x_1)^2 + (y-x_2)^2 + (z-x_3)^2)^{-\frac{3}{2}} \right] =$$

$$= (x-x_1) \cdot \left(-\frac{3}{2}\right) \cdot \|\vec{x}-\vec{r}\|^{-5} \cdot (2z-2x_3) - (z-x_3) \cdot \left(-\frac{3}{2}\right) \cdot \|\vec{x}-\vec{r}\|^{-5} \cdot (2x-2x_1) =$$

$$= -3 \cdot \|\vec{x}-\vec{r}\|^{-5} \cdot \underbrace{[(x-x_1) \cdot (z-x_3) - (z-x_3) \cdot (x-x_1)]}_{\vec{0}} = \underline{\underline{0}}$$

$$A_3 = \frac{\partial}{\partial x} \left[(y-x_2) \cdot ((x-x_1)^2 + (y-x_2)^2 + (z-x_3)^2)^{-\frac{3}{2}} \right] - \frac{\partial}{\partial y} \left[(x-x_1) \cdot ((x-x_1)^2 + (y-x_2)^2 + (z-x_3)^2)^{-\frac{3}{2}} \right] =$$

$$= (y-x_2) \cdot \left(-\frac{3}{2}\right) \cdot \|\vec{x}-\vec{r}\|^{-5} \cdot (2x-2x_1) - (x-x_1) \cdot \left(-\frac{3}{2}\right) \cdot \|\vec{x}-\vec{r}\|^{-5} \cdot (2y-2x_2) =$$

$$= -3 \cdot \|\vec{x}-\vec{r}\|^{-5} \cdot \underbrace{[(y-x_2) \cdot (x-x_1) - (x-x_1) \cdot (y-x_2)]}_{\vec{0}} = \underline{\underline{0}}$$

$$\vec{0} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \underline{\underline{\vec{0}}}$$

(Macht natürlich Sinn, da der Vektor $(\vec{r}-\vec{x})$ ja nur verkürzt wird. Also muss Kreuzprodukt immer noch Null sein.)